Path A - Attention Computed by Hand

Every number below was produced by `path_a_walkthrough.py` (NumPy, no deep-learning framework): a 3-token toy sequence with D=8, d_k=d_v=4.

0. Setup and shapes

Symbol	Shape	Meaning
X	(3, 8)	token embeddings
PE	(3, 4)	sinusoidal positions (broadcast over dims)
Xin	(3, 8)	X + PE
Q,K	(3, 4)	queries and keys
V	(3, 4)	values
S	(3, 3)	raw scores Q K^T
Sn = S / sqrt(d_k)	(3, 3)	scaled scores, sqrt(4)=2
A	(3, 3)	softmax weights
Z	(3, 4)	context vectors A V